

## Section 1. Multiplication of Polynomials I

*Multiply the polynomials and simplify your answer.*

$$(1) \quad 2a(a + 4) =$$

$$(2) \quad 2a(a + 9) =$$

$$(3) \quad (x + 1)(x + 1) =$$

$$(4) \quad 3a(a - 9) =$$

$$(5) \quad 5a(a + 1) =$$

$$(6) \quad (y + 3)(y - 2) =$$

## Section 1. Multiplication of Polynomials I

*Multiply the polynomials and simplify your answer.*

$$(1) \quad 3x(x - 7) =$$

$$(2) \quad 4x(x - 1) =$$

$$(3) \quad (y - 1)(y + 2) =$$

$$(4) \quad (y - 4)(y + 2) =$$

$$(5) \quad 4a(a - 7) =$$

$$(6) \quad (x - 2)(x + 5) =$$

## Section 1. Multiplication of Polynomials I

*Multiply the polynomials and simplify your answer.*

$$(1) \quad 2x(x - 3) =$$

$$(2) \quad 5y(y - 8) =$$

$$(3) \quad 2a(a - 5) =$$

$$(4) \quad 4y(y + 8) =$$

$$(5) \quad (x - 1)(x + 6) =$$

$$(6) \quad 4x(x + 9) =$$

## Section 1. Multiplication of Polynomials I

*Multiply the polynomials and simplify your answer.*

$$(1) \quad 2y(y - 6) =$$

$$(2) \quad (x + 4)(x - 5) =$$

$$(3) \quad 4a(a + 6) =$$

$$(4) \quad 2a(a + 5) =$$

$$(5) \quad 2y(y - 8) =$$

$$(6) \quad 4y(y + 5) =$$

## Section 1. Multiplication of Polynomials I

*Multiply the polynomials and simplify your answer.*

$$(1) \quad (y - 1)(y - 1) =$$

$$(2) \quad 2x(x + 2) =$$

$$(3) \quad (x - 2)(x + 2) =$$

$$(4) \quad (y + 2)(y - 5) =$$

$$(5) \quad (y - 4)(y + 5) =$$

$$(6) \quad 3x(x + 1) =$$

## Section 1. Multiplication of Polynomials I

*Multiply the polynomials and simplify your answer.*

$$(1) \quad 3x(x + 8) =$$

$$(2) \quad (x + 3)(x + 4) =$$

$$(3) \quad (x + 1)(x + 6) =$$

$$(4) \quad 3y(y + 8) =$$

$$(5) \quad (x + 2)(x - 4) =$$

$$(6) \quad (y - 3)(y + 4) =$$

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*Multiply the polynomials and simplify your answer.*

$$(1) \quad 4x(x + 9) =$$

$$(2) \quad (x - 1)(x + 5) =$$

$$(3) \quad 2x(x + 2) =$$

$$(4) \quad (x + 2)(x + 5) =$$

$$(5) \quad (y - 3)(y + 2) =$$

$$(6) \quad (y - 2)(y - 6) =$$

## Section 1. Multiplication of Polynomials I

*Multiply the polynomials and simplify your answer.*

$$(1) \quad 2x(x + 1) =$$

$$(2) \quad 2y(y + 9) =$$

$$(3) \quad (y + 2)(y - 5) =$$

$$(4) \quad 5y(y + 2) =$$

$$(5) \quad (y - 4)(y + 4) =$$

$$(6) \quad 2y(y + 6) =$$

## Section 1. Multiplication of Polynomials I

*Multiply the polynomials and simplify your answer.*

$$(1) \quad (x + 1)(x + 4) =$$

$$(2) \quad 4x(x + 2) =$$

$$(3) \quad (y - 2)(y + 4) =$$

$$(4) \quad (x + 2)(x + 3) =$$

$$(5) \quad 4y(y - 4) =$$

$$(6) \quad 4a(a + 5) =$$

## Section 1. Multiplication of Polynomials I

*Multiply the polynomials and simplify your answer.*

$$(1) \quad (x - 2)(x - 6) =$$

$$(2) \quad 4x(x - 2) =$$

$$(3) \quad 5x(x - 8) =$$

$$(4) \quad (x + 2)(x - 1) =$$

$$(5) \quad (y - 1)(y - 3) =$$

$$(6) \quad (x + 1)(x - 3) =$$

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*Multiply the polynomials and simplify your answer.*

$$(1) \quad 2a(a - 6) =$$

$$(2) \quad (x + 4)(x - 5) =$$

$$(3) \quad 5x(x + 9) =$$

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*Multiply the polynomials and simplify your answer.*

$$(1) \quad 5y(y - 1) =$$

$$(2) \quad 4x(x - 8) =$$

$$(3) \quad 2x(x + 3) =$$

$$(4) \quad (x + 4)(x - 2) =$$

$$(5) \quad (x + 3)(x + 2) =$$

$$(6) \quad (y + 2)(y + 2) =$$

## Section 1. Multiplication of Polynomials I

*Multiply the polynomials and simplify your answer.*

$$(1) \quad 4y(y - 1) =$$

$$(2) \quad (x - 4)(x - 5) =$$

$$(3) \quad (y - 2)(y + 5) =$$

$$(4) \quad 4a(a + 6) =$$

$$(5) \quad 4x(x + 3) =$$

$$(6) \quad 3y(y - 2) =$$

## Section 1. Multiplication of Polynomials I

*Multiply the polynomials and simplify your answer.*

$$(1) \quad 4y(y + 9) =$$

$$(2) \quad 3x(x + 3) =$$

$$(3) \quad 5y(y - 9) =$$

$$(4) \quad (x - 2)(x + 4) =$$

$$(5) \quad (y - 1)(y - 3) =$$

$$(6) \quad (x - 4)(x - 2) =$$

## Section 1. Multiplication of Polynomials I

*Multiply the polynomials and simplify your answer.*

$$(1) \quad (x - 4)(x + 3) =$$

$$(2) \quad (x + 1)(x + 4) =$$

$$(3) \quad (x + 4)(x - 2) =$$

$$(4) \quad (y - 2)(y - 4) =$$

$$(5) \quad (y + 4)(y - 3) =$$

$$(6) \quad 2y(y + 2) =$$

## Section 1. Multiplication of Polynomials I

*Multiply the polynomials and simplify your answer.*

$$(1) \quad (x - 4)(x + 4) =$$

$$(2) \quad 2x(x + 4) =$$

$$(3) \quad (x - 2)(x + 1) =$$

$$(4) \quad (y - 4)(y - 5) =$$

$$(5) \quad 5y(y - 4) =$$

$$(6) \quad (x + 3)(x - 4) =$$

## Section 1. Multiplication of Polynomials I

*Multiply the polynomials and simplify your answer.*

$$(1) \quad 3x(x + 3) =$$

$$(2) \quad 5a(a - 4) =$$

$$(3) \quad (y + 2)(y + 5) =$$

$$(4) \quad (x - 1)(x - 3) =$$

$$(5) \quad 2a(a + 4) =$$

$$(6) \quad (x - 1)(x - 3) =$$

## Section 1. Multiplication of Polynomials I

*Multiply the polynomials and simplify your answer.*

$$(1) \quad (y - 3)(y + 3) =$$

$$(2) \quad 2x(x - 7) =$$

$$(3) \quad 5a(a - 8) =$$

$$(4) \quad 2y(y + 7) =$$

$$(5) \quad (y - 4)(y - 5) =$$

$$(6) \quad (y - 4)(y - 3) =$$

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*Multiply the polynomials and simplify your answer.*

$$(1) \quad 2a(a + 6) =$$

$$(2) \quad 3x(x - 8) =$$

$$(3) \quad (x + 2)(x - 3) =$$

$$(4) \quad 4x(x + 9) =$$

$$(5) \quad (x - 1)(x + 5) =$$

$$(6) \quad (x - 1)(x - 5) =$$

## Section 1. Multiplication of Polynomials I

*Multiply the polynomials and simplify your answer.*

$$(1) \quad 4x(x + 8) =$$

$$(2) \quad (y + 4)(y - 6) =$$

$$(3) \quad 4a(a - 8) =$$

$$(4) \quad (y + 3)(y - 2) =$$

$$(5) \quad (x - 4)(x - 5) =$$

$$(6) \quad 5y(y + 8) =$$

## Answer Key

### JPMethod Polynomials

#### F1a

- (1)  $2a^2 + 8a$
- (2)  $2a^2 + 18a$
- (3)  $x^2 + 2x + 1$
- (4)  $3a^2 - 27a$
- (5)  $5a^2 + 5a$
- (6)  $y^2 + y - 6$

#### F2a

- (1)  $3x^2 - 21x$
- (2)  $4x^2 - 4x$
- (3)  $y^2 + y - 2$
- (4)  $y^2 - 2y - 8$
- (5)  $4a^2 - 28a$
- (6)  $x^2 + 3x - 10$

#### F1b

- (1)  $2y^2 - 12y$
- (2)  $x^2 - x - 20$
- (3)  $4a^2 + 24a$
- (4)  $2a^2 + 10a$
- (5)  $2y^2 - 16y$
- (6)  $4y^2 + 20y$

#### F2b

- (1)  $2x^2 - 6x$
- (2)  $5y^2 - 40y$
- (3)  $2a^2 - 10a$
- (4)  $4y^2 + 32y$
- (5)  $x^2 + 5x - 6$
- (6)  $4x^2 + 36x$

## Answer Key

### JPMethod Polynomials

#### F5a

- (1)  $x^2 + 5x + 4$
- (2)  $4x^2 + 8x$
- (3)  $y^2 + 2y - 8$
- (4)  $x^2 + 5x + 6$
- (5)  $4y^2 - 16y$
- (6)  $4a^2 + 20a$

#### F6a

- (1)  $x^2 - 8x + 12$
- (2)  $4x^2 - 8x$
- (3)  $5x^2 - 40x$
- (4)  $x^2 + x - 2$
- (5)  $y^2 - 4y + 3$
- (6)  $x^2 - 2x - 3$

#### F5b

- (1)  $5y^2 - 5y$
- (2)  $4x^2 - 32x$
- (3)  $2x^2 + 6x$
- (4)  $x^2 + 2x - 8$
- (5)  $x^2 + 5x + 6$
- (6)  $y^2 + 4y + 4$

#### F6b

- (1)  $2a^2 - 12a$
- (2)  $x^2 - x - 20$
- (3)  $5x^2 + 45x$
- (4)  $y^2 - 7y + 6$
- (5)  $4a^2 - 28a$
- (6)  $y^2 - 4y + 4$

## Answer Key

### JPMethod Polynomials

#### F7a

- (1)  $4y^2 - 4y$
- (2)  $x^2 - 9x + 20$
- (3)  $y^2 + 3y - 10$
- (4)  $4a^2 + 24a$
- (5)  $4x^2 + 12x$
- (6)  $3y^2 - 6y$

#### F8a

- (1)  $4y^2 + 36y$
- (2)  $3x^2 + 9x$
- (3)  $5y^2 - 45y$
- (4)  $x^2 + 2x - 8$
- (5)  $y^2 - 4y + 3$
- (6)  $x^2 - 6x + 8$

#### F7b

- (1)  $x^2 - 16$
- (2)  $2x^2 + 8x$
- (3)  $x^2 - x - 2$
- (4)  $y^2 - 9y + 20$
- (5)  $5y^2 - 20y$
- (6)  $x^2 - x - 12$

#### F8b

- (1)  $x^2 - x - 12$
- (2)  $x^2 + 5x + 4$
- (3)  $x^2 + 2x - 8$
- (4)  $y^2 - 6y + 8$
- (5)  $y^2 + y - 12$
- (6)  $2y^2 + 4y$

## Answer Key

### JPMethod Polynomials

#### F3a

- (1)  $y^2 - 2y + 1$
- (2)  $2x^2 + 4x$
- (3)  $x^2 - 4$
- (4)  $y^2 - 3y - 10$
- (5)  $y^2 + y - 20$
- (6)  $3x^2 + 3x$

#### F4a

- (1)  $3x^2 + 24x$
- (2)  $x^2 + 7x + 12$
- (3)  $x^2 + 7x + 6$
- (4)  $3y^2 + 24y$
- (5)  $x^2 - 2x - 8$
- (6)  $y^2 + y - 12$

#### F3b

- (1)  $2x^2 + 2x$
- (2)  $2y^2 + 18y$
- (3)  $y^2 - 3y - 10$
- (4)  $5y^2 + 10y$
- (5)  $y^2 - 16$
- (6)  $2y^2 + 12y$

#### F4b

- (1)  $4x^2 + 36x$
- (2)  $x^2 + 4x - 5$
- (3)  $2x^2 + 4x$
- (4)  $x^2 + 7x + 10$
- (5)  $y^2 - y - 6$
- (6)  $y^2 - 8y + 12$

### F9a

- (1)  $3x^2 + 9x$
- (2)  $5a^2 - 20a$
- (3)  $y^2 + 7y + 10$
- (4)  $x^2 - 4x + 3$
- (5)  $2a^2 + 8a$
- (6)  $x^2 - 4x + 3$

### F10a

- (1)  $y^2 - 9$
- (2)  $2x^2 - 14x$
- (3)  $5a^2 - 40a$
- (4)  $2y^2 + 14y$
- (5)  $y^2 - 9y + 20$
- (6)  $y^2 - 7y + 12$

### F9b

- (1)  $4x^2 + 32x$
- (2)  $y^2 - 2y - 24$
- (3)  $4a^2 - 32a$
- (4)  $y^2 + y - 6$
- (5)  $x^2 - 9x + 20$
- (6)  $5y^2 + 40y$

### F10b

- (1)  $2a^2 + 12a$
- (2)  $3x^2 - 24x$
- (3)  $x^2 - x - 6$
- (4)  $4x^2 + 36x$
- (5)  $x^2 + 4x - 5$
- (6)  $x^2 - 6x + 5$

